

27 Ans: D

Option A is incorrect because photoelectric effect can occur even for very low intensity but sufficiently high frequency radiation (above the threshold frequency f_0).

Option B is incorrect because photoelectrons are emitted only if the frequency of radiation is greater than a minimum frequency known as threshold frequency f_0 ; since $\lambda_0 = \frac{c}{f_0}$, the wavelength of the radiation must be smaller than the threshold wavelength, which is instead a maximum value.

Option C is incorrect because the maximum kinetic energy and thus speed of the photoelectrons is independent of intensity but depends on the frequency of radiation and work function of the metal, as given by $hf = \Phi + E_{k,max}$.

Option D is correct since $I = \frac{N_P}{t} \frac{hf}{A}$, at constant intensity and increased frequency, the rate of photons incident on the metal decreases, the lesser the rate of emission of photoelectrons.

28 Ans: A